Restaurant Management System

Abdulrahman Musab Hassan

Ali Zaid Mohammed

Rand Saad Abdul Sattar

Abdullah Yassar Turki

University of Baghdad | College of AI | Department of Big Data

Supervised by: Dr. Suhad Faisal

INTRODUCTION

Database Project Overview

What Is This System?

This project presents a **Restaurant Management System** — a relational database designed to manage customers, meals, and orders efficiently in a restaurant environment.

The 5 Main Tables

01

Categories

Organizes meal types such as appetizers, main courses, desserts, and drinks.

02

Customers

Stores customer personal information and contact details.

03

Meals

Lists all available meals with prices, descriptions, and category links.

04

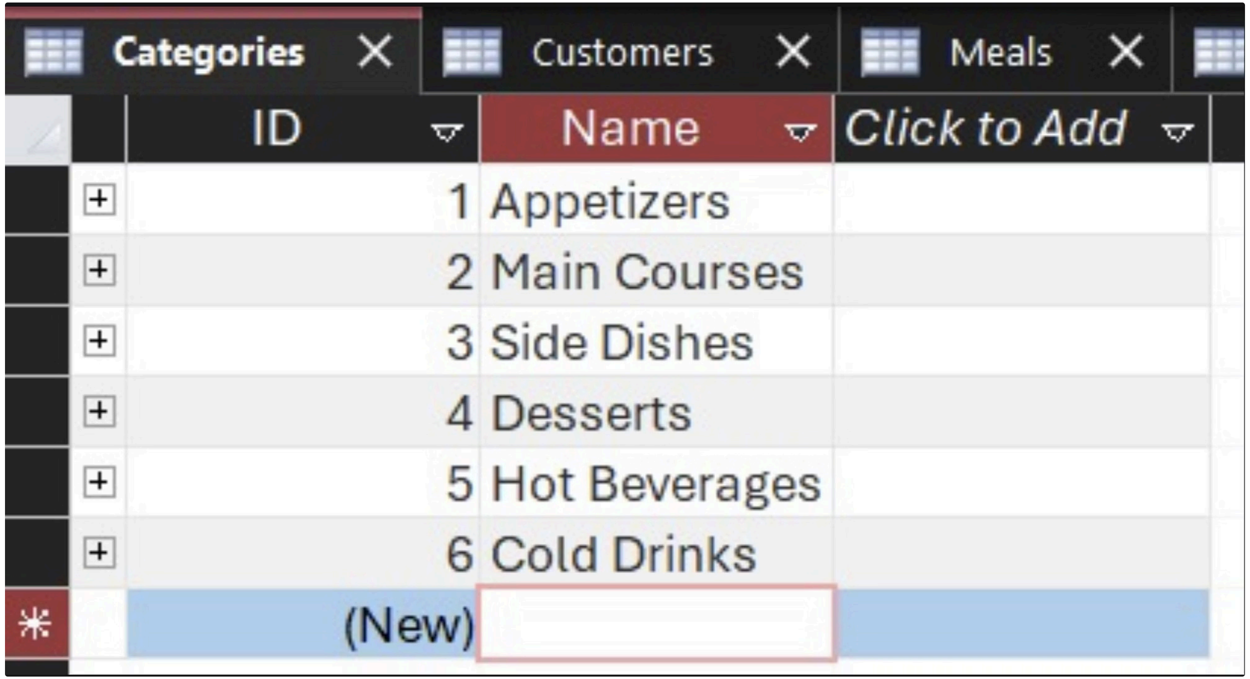
Orders

Tracks each order placed, its date, total amount, and current status.

05

Order Details

Connects orders with specific meals and records quantities ordered.



Categories | Customers | Meals

ID	Name	Click to Add
1	Appetizers	
2	Main Courses	
3	Side Dishes	
4	Desserts	
5	Hot Beverages	
6	Cold Drinks	
(New)		

TABLE 1

Categories Table

Purpose

Contains different types of meals available in the restaurant, such as appetizers, main courses, desserts, and drinks.

Key Field

Each category is assigned a **unique Category ID**, which serves as the primary key and is used to link meals to their respective category.

Role in the System

The Categories table helps organize the meal catalog and enables efficient filtering and browsing of the restaurant's offerings.

Categories | Customers | Meals | Order_Details | Orders

ID	Customer_N	Phone	Adress	Click to Add
1	Zainab Mahmo	7612223344	Adhamiya, Bag	
2	Mohammed Qa	7023334455	Ghazaliya, Bag	
3	Fatima Hussei	7604445566	Palestine Stree	
4	Yassir Ammar	7205556677	Dora, Baghdad	
5	Rawan Fares	7306667788	Amriya, Baghd	
6	Hussein Ali	7207778899	Bayaa, Baghda	
7	Huda Kareem	7118889900	Shaab, Baghda	
8	Abbas Jassim	7119990011	Kadhimiyah, Ba	
9	Suha Nabeel	7011234567	Karkh, Baghda	
10	Bilal Haider	7122345678	Saidiya, Baghd	
11	Ruqaya Saad	7433456789	Shuala, Baghd	
12	Saif Salman	7244567890	Rusafa, Baghd	
13	Maryam Walid	7355678901	Hurriya, Baghd	
14	Laith Hassan	7066789012	Zaafaraniya, Ba	
15	Dania Tariq	7577890123	Qadisiyah, Bag	
16	Ali Youssef	7645612398	Zayouna, Bagh	
17	Muna Tariq	7156723409	Karrada, Bagh	
18	Hassan Jabbar	7367834510	Mansour, Bagh	
19	Laila Majeed	72789456210	Yarmouk, Bagh	
20	Hasan Tariq	76890567320	Jadriya, Baghd	
(New)		0		

TABLE 2

Customers Table

Purpose

Stores all relevant customer information required for managing orders and maintaining relationships.

Stored Fields

Each record includes the customer's **name**, **phone number**, and **address**, along with a unique Customer ID as the primary key.

Role in the System

The Customer ID is referenced in the Orders table to link every order back to the customer who placed it, ensuring data integrity across the system.

Categories | Customers | Meals | Order_Details | Orders | Query1

ID	Meal name	Price	Description	Category_ID	Click to Add
1	Hummus	4000	Mashed chickp	1	
2	Baba Ghanous	4500	Roasted eggpla	1	
3	Lentil Soup	3000	Warm and corr	1	
4	Cheese Sambo	5000	Crispy fried pas	1	
5	Chicken Tikka	12000	2 skewers of gr	2	
6	Beef Shawarm	7000	Sliced beef witl	2	
7	Iraqi Kebab	14000	3 skewers of gr	2	
8	Margherita Pizz	10000	Classic pizza w	2	
9	Crispy Chicken	9000	Fried chicken b	2	
10	French Fries	3000	Crispy golden f	3	
11	Garlic Bread	4000	Toasted bague	3	
12	Spicy Potato	4500	Fried potato cu	3	
13	Chocolate Cak	6000	Rich chocolate	4	
14	Kunafa	7000	Warm sweet pa	4	
15	Rice Pudding	3500	Creamy milk ar	4	
16	Turkish Coffee	3000	Dark roasted tl	5	
17	Iraqi Tea (Chai)	1000	Strong black te	5	
18	Fresh Lemonac	4000	Freshly squeez	6	
19	Pepsi	1500	Chilled canned	6	
20	Mineral Water	1000	Cold bottled w	6	
(New)		0		0	

TABLE 3

Meals Table

Purpose

Lists all meals available in the restaurant, serving as the central product catalog of the database.

Stored Fields

Each meal record contains a **name**, **price**, **description**, and a **Category ID** that links the meal to its corresponding category.

Role in the System

The Meal ID from this table is referenced in the Order Details table, connecting every ordered item back to its full meal record.

Categories | Customers | Meals | Order_Details

ID	Order_ID	Meal_ID	Quantity	Click to Add
1	1	5	1	
2	1	6	1	
3	1	13	1	
4	2	7	1	
5	3	8	2	
6	3	19	3	
7	4	1	1	
8	4	9	1	
9	5	10	2	
10	5	18	2	
11	6	7	2	
12	6	16	2	
13	7	14	3	
14	8	2	1	
15	8	11	1	
16	9	6	2	
17	9	19	2	
18	10	7	1	
19	10	17	2	
20	11	8	1	
21	11	10	1	
22	12	5	2	
23	12	1	1	
24	13	9	1	
25	13	19	1	
26	13	13	1	
27	14	14	2	
28	14	16	2	
29	15	3	2	
30	15	11	1	
31	16	12	1	
32	16	18	1	
33	17	7	3	
34	17	15	3	
35	18	6	1	
36	18	10	1	
37	18	19	1	
38	19	4	2	
39	19	17	2	
40	20	8	2	
41	20	18	2	
42	20	13	1	
(New)	0	0	0	

TABLE 4

Order Details Table

Purpose

Records exactly what was ordered within each individual order, acting as a junction table between Orders and Meals.

Stored Fields

Each record contains an **Order ID**, a **Meal ID**, and the **quantity** ordered, precisely capturing the items in every transaction.

Role in the System

This table is the bridge that connects the Orders table with the Meals table, enabling the system to calculate totals and generate detailed receipts.

Categories | Customers | Meals | Order_Details | Orders | Query1

ID	Customer_I	Order_Date	Total_Amou	Status	Click to Add
1	1	3/28/2026 13:1	25000	Delivered	
2	8	3/28/2026 14:0	14000	Delivered	
3	12	3/28/2026 18:3	24500	Delivered	
4	3	3/29/2026 12:4	13000	Delivered	
5	15	3/29/2026 13:2	14000	Cancelled	
6	2	3/29/2026 19:1	34000	Delivered	
7	19	3/29/2026 20:0	21000	Delivered	
8	7	3/30/2026 13:0	8500	Delivered	
9	10	3/30/2026 14:2	17000	Delivered	
10	1	3/30/2026 15:1	16000	Delivered	
11	4	3/30/2026 18:4	13000	On the way	
12	5	3/30/2026 19:1	28000	On the way	
13	6	3/30/2026 19:3	16500	Preparing	
14	9	3/30/2026 19:4	20000	Preparing	
15	11	3/30/2026 20:0	10000	Preparing	
16	13	3/30/2026 20:1	8500	Preparing	
17	14	3/30/2026 20:3	52500	Cancelled	
18	16	3/30/2026 21:0	11500	Preparing	
19	17	3/30/2026 21:2	12000	On the way	
20	20	3/30/2026 22:1	34000	Preparing	
(New)	0				

TABLE 5

Orders Table

Purpose

Stores high-level information about each order placed in the restaurant system.

Stored Fields

Each record holds the **Customer ID**, **order date**, **total amount**, and **status** — indicating whether the order is *delivered, preparing,* or *on the way.*

Role in the System

The Orders table links customers to their transactions and coordinates with the Order Details table to present a complete view of every order lifecycle.

QUERY RESULT

Final Query & Results

What the Query Does

This final query combines data from multiple tables to produce a unified, human-readable result showing exactly what each customer ordered.

Customer Name

Identifies who placed the order

Order Date

When the order was placed

Meal Name & Quantity

What was ordered and how much

Why This Query Matters

By joining the Customers, Orders, Order Details, and Meals tables, the query provides a **clear and complete picture** of every transaction — making it easy to analyze customer behavior and restaurant performance.

This multi-table JOIN is the core analytical query of the system, demonstrating the relational power of the database design.

Categories | Customers | Meals

Customer_N	Order_Date	Meal name	Quantity	Price
Zainab Mahmc	3/28/2026 13:	Chicken Tikka	1	12000
Zainab Mahmc	3/28/2026 13:	Beef Shawarma	1	7000
Zainab Mahmc	3/28/2026 13:	Chocolate Cak	1	6000
Abbas Jassim	3/28/2026 14:	Iraqi Kebab	1	14000
Saif Salman	3/28/2026 18:	Margherita Piz	2	10000
Saif Salman	3/28/2026 18:	Pepsi	3	1500
Fatima Hussei	3/29/2026 12:	Hummus	1	4000
Fatima Hussei	3/29/2026 12:	Crispy Chicken	1	9000
Dania Tariq	3/29/2026 13:	French Fries	2	3000
Dania Tariq	3/29/2026 13:	Fresh Lemonac	2	4000
Mohammed Qa	3/29/2026 19:	Iraqi Kebab	2	14000
Mohammed Qa	3/29/2026 19:	Turkish Coffee	2	3000
Laila Majeed	3/29/2026 20:	Kunafa	3	7000
Huda Kareem	3/30/2026 13:	Baba Ghanous	1	4500
Huda Kareem	3/30/2026 13:	Garlic Bread	1	4000
Bilal Haider	3/30/2026 14:	Beef Shawarma	2	7000
Bilal Haider	3/30/2026 14:	Pepsi	2	1500
Zainab Mahmc	3/30/2026 15:	Iraqi Kebab	1	14000
Zainab Mahmc	3/30/2026 15:	Iraqi Tea (Chai	2	1000
Yassir Ammar	3/30/2026 18:	Margherita Piz	1	10000
Yassir Ammar	3/30/2026 18:	French Fries	1	3000
Rawan Fares	3/30/2026 19:	Chicken Tikka	2	12000
Rawan Fares	3/30/2026 19:	Hummus	1	4000
Hussein Ali	3/30/2026 19:	Crispy Chicken	1	9000
Hussein Ali	3/30/2026 19:	Pepsi	1	1500
Hussein Ali	3/30/2026 19:	Chocolate Cak	1	6000
Suha Nabeel	3/30/2026 19:	Kunafa	2	7000
Suha Nabeel	3/30/2026 19:	Turkish Coffee	2	3000
Ruqaya Saad	3/30/2026 20:	Lentil Soup	2	3000
Ruqaya Saad	3/30/2026 20:	Garlic Bread	1	4000
Maryam Walic	3/30/2026 20:	Spicy Potato	1	4500
Maryam Walic	3/30/2026 20:	Fresh Lemonac	1	4000
Laith Hassan	3/30/2026 20:	Iraqi Kebab	3	14000
Laith Hassan	3/30/2026 20:	Rice Pudding	3	3500
Ali Youssef	3/30/2026 21:	Beef Shawarma	1	7000
Ali Youssef	3/30/2026 21:	French Fries	1	3000
Ali Youssef	3/30/2026 21:	Pepsi	1	1500
Muna Tariq	3/30/2026 21:	Cheese Sambo	2	5000
Muna Tariq	3/30/2026 21:	Iraqi Tea (Chai	2	1000
Hasan Tariq	3/30/2026 22:	Margherita Piz	2	10000
Hasan Tariq	3/30/2026 22:	Fresh Lemonac	2	4000
Hasan Tariq	3/30/2026 22:	Chocolate Cak	1	6000

CONCLUSION

Conclusion

1

Organize Data

Structure restaurant information across 5 well-defined relational tables

2

Track Orders

Monitor every order from placement through delivery with status tracking

3

Improve Service

Enable data-driven decisions to enhance customer satisfaction and efficiency

Key Takeaways

A 5-table relational database successfully models all core restaurant operations

Foreign key relationships between tables ensure data consistency and integrity

The final multi-table query provides clear, actionable insights into customer orders

Thank You

Questions and Discussion